

SAHAAL MK

Chennai, India | GitHub: github.com/sahaalmkjee-droid | Kaggle: [sahaalmkjee](https://kaggle.com/sahaalmkjee)

EDUCATION

Indian Institute of Technology (IIT) Madras

B.Tech (Undergraduate) | Competitive Programming: Codeforces 910

Chennai, India

2025 – Present

TECHNICAL SKILLS

- **Languages:** Python, C++, SQL, JavaScript, HTML5/CSS3
- **Frameworks & Tools:** FastAPI, React, PyTorch, PyTorch Lightning, scikit-learn, OpenCV, YOLOv8, Docker, Git
- **Data & AI Architecture:** PostgreSQL, pgvector, Vector Search, CNNs, Transformers, Feature Engineering, REST APIs
- **Developer Practices:** CI/CD, Asynchronous Programming, Microservices, Agile Workflow

KEY TECHNICAL PROJECTS

Autonomous Career Intelligence Agent (NEXUS)

FastAPI, React, pgvector, Gemini API

- Built an autonomous career pipeline scraping job listings using Playwright and parsing unstructured text into strict JSON schemas via Gemini 2.5 Flash.
- Implemented semantic vector matching with pgvector (cosine distance) and built a tool-calling chat agent using function calling.
- Created an asynchronous video/audio briefing engine with non-blocking lifecycle status polling.

End-to-End Real-Time Toxicity Classification Pipeline

Python, FastAPI, React, Hugging Face

- Designed and integrated a real-time automated text toxicity moderation app connecting a React client to a Python FastAPI backend.
- Deployed BERT transformer models for low-latency inference on incoming user streams.

Stacked Neural Network & Ensembling Pipeline

PyTorch, CatBoost, Scikit-Learn

- Engineered a stacked deep learning ensemble combining multi-head neural networks with a gradient boosted meta-learner.
- Secured 22nd place globally in a competitive data challenge through rigorous cross-validation and hyperparameter tuning.

Automated Computer Vision Detection Systems

YOLOv8, PyTorch, OpenCV

- Fine-tuned YOLOv8 bounding-box models on custom video feeds with real-time tracking algorithms.
- Automated dataset annotation, bounding box calculations, and GPU training acceleration routines.

LEADERSHIP & RECOGNITION

- **Global Competitive Machine Learning:** Ranked 22nd globally on Kaggle challenge leaderboard.
- **Project Point of Contact (She-ViL, IIT Madras):** Directed STEAM training workshops, technical modules, and student tenures.
- **Junior Executive, Tensors NGO:** Built technical task coordination pipelines and student community initiatives.